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ENBC 332

Homework 6 Discussion and Outputs

Problem 1:

1. The null hypothesis for this question average daily sodium intake for a specific population is 2,300mg. The alternative hypothesis is that the average daily sodium intake is either greater than or less than 2,300mg.
2. The p-value is approximately 0.473, which indicates there is a 47.3% chance the average daily sodium intake for the population is 2,300mg. Additionally, the p-value is greater than 0.1 which is within the no evidence range showing that it is not statistically significant.
3. The chosen significance level was 0.05 or 5%, and the p-value is greater than it. This results in a failure to reject the null hypothesis, meaning there is not enough evidence to prove the nutritionist's claim.

Problem 2:

1. The null hypothesis would be a new cholesterol lowering drug that reduces the average LDL cholesterol level in adults greater than or equal to 120mg/dl after 8 weeks of treatment. The alternative hypothesis is that the new cholesterol lowering drug reduces the average LDL cholesterol level in adults less than 120mg/dl after 8 weeks of treatment.
2. The p-value is approximately 0, which is inside the very strong range of statistical significance and indicates that the observed sample mean is highly statistically significant.
3. The chosen significance level was 0.05 or 5%, and the p-value is less than it. This results in a success in rejecting the null hypothesis, meaning that the claim from the pharmaceutical company is valid and true.

```
> # Variables
> mean_pop <- 2300
> n <- 30
> intake_data <- c(2100, 2500, 2300, 1950, 2600, 2200, 2450, 2150, 2400, 2550,
+                  2350, 2250, 2050, 2650, 2300, 2400, 2100, 2500, 2350, 2200,
+                  2450, 2150, 2250, 2550, 2050, 2600, 2300, 2150, 2400, 2500)
> mean_sam <- mean(intake_data)
> sd_sam <- sd(intake_data)
> # Computations
> t_stat <- (mean_sam - mean_pop) / (sd_sam / sqrt(n))
> p_value_t <- 2 * (1 - pt(abs(t_stat), df = n-1))
> cat("T-test:", "\nT-Statistic:", t_stat, "\nP-Value:", p_value_t)
T-test:
T-Statistic: 0.7270172
P-Value: 0.4730439
> # Variables
> mean_pop <- 120
> n <- 25
> ldl_data <- c(110, 112, 115, 113, 111, 114, 109, 112, 111, 115, 113, 110,
+              112, 109, 111, 113, 114, 110, 112, 111, 109, 113, 112, 110, 111)
> mean_sam <- mean(ldl_data)
> sd_sam <- sd(ldl_data)
> # Computations
> t_stat <- (mean_sam - mean_pop) / (sd_sam / sqrt(n))
> p_value_t <- (1 - pt(abs(t_stat), df = n-1))
> cat("T-test:", "\nT-Statistic:", t_stat, "\nP-Value:", p_value_t)
T-test:
T-Statistic: -23.46379
P-Value: 0
> |
```